

GUARD: Geometric Uncertainty-Aware Robust Denoiser for Point Cloud Segmentation

Supplementary Material

6. Per-Category mIoU Results

We report the per-category mIoU on ShapeNetPart to complement the main results in Table 2. As shown in the table above, our method achieves consistent improvements over the vanilla PointNet++ baseline across nearly all categories. Notably, the performance gains are more pronounced on geometrically complex or fine-grained objects such as *Cap*, *Skateboard*, *Chair*, and *Knife*, indicating that our geometric uncertainty modeling enhances feature reliability for both smooth and intricate structures. Meanwhile, categories with large planar or symmetric parts (e.g., *Table*, *Lamp*) also benefit from the hybrid denoising process, maintaining stable segmentation even under high corruption severity. Overall, these per-category results further validate the robustness and generality of our approach across diverse object geometries.

cade counterpart, confirming its practical advantage for efficient denoising in robotic perception tasks.

7. Comparison with Fusion and Cascade Strategies

To further assess the efficiency and robustness of our uncertainty integration design, we compare the proposed *Fusion* strategy with a two-stage *Cascade* baseline. The *Cascade* method sequentially applies two kinds of uncertainty filters, while the *Fusion* approach combines them in a single step via a weighted sum, reducing computational overhead and improving stability.

Table 7. Comparison between Fusion and Cascade strategies. mIoU measures segmentation accuracy, Noise Ratio (%) denotes the proportion of residual noisy points, and mean latency (ms) reflects average inference time per sample.

Method	mIoU $\uparrow$	Noise Ratio (%) $\downarrow$	Mean Latency (ms) $\downarrow$
Cascade	0.6282	24.41	189.68
Fusion (ours)	0.7240	22.30	169.51

The results show that the *Fusion* design improves segmentation performance by reducing the retained noise ratio and achieving a lower mean latency compared to the *Cas-*

Table 6. Per-category mIoU results on ShapeNetPart for different methods. We report results for PointNet++ and DGCNN backbones under three settings: Vanilla, PointCVaR, and our GUARD.

Backbone	Method	Per-Category mIoU $\uparrow$ on ShapeNetPart															
		Earphone	Motorbike	Rocket	Car	Laptop	Cap	Skateboard	Mug	Guitar	Bag	Lamp	Table	Airplane	Pistol	Chair	Knife
PointNet++	Vanilla	0.4416	0.4975	0.4239	0.4732	0.7061	0.5310	0.3996	0.6646	0.7339	0.5384	0.7030	0.5404	0.5975	0.6010	0.6621	0.6251
	+PointCVaR	0.4631	0.3115	0.3826	0.5128	0.7373	0.5387	0.3983	0.5689	0.7278	0.4252	0.6480	0.5763	0.6045	0.5767	0.6446	0.5848
	+GUARD	0.5708	0.5186	0.4322	0.5181	0.7874	0.6353	0.5370	0.6471	0.8131	0.4990	0.7284	0.6591	0.7086	0.6005	0.7739	0.7449
DGCNN	Vanilla	0.5222	0.3232	0.3574	0.4599	0.7023	0.5664	0.3668	0.5808	0.7290	0.5617	0.6975	0.6039	0.6075	0.5243	0.7100	0.6067
	+PointCVaR	0.5762	0.3343	0.3791	0.4470	0.7674	0.6254	0.3730	0.5101	0.7526	0.6092	0.6928	0.6385	0.6398	0.5568	0.7364	0.6663
	+GUARD	0.5882	0.4085	0.4135	0.3786	0.8309	0.6634	0.5881	0.7499	0.8421	0.6325	0.7598	0.7319	0.6826	0.6643	0.8331	0.7203